

QINHE PENG

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Education

Tsinghua University (THU), Beijing, China

09/2020-06/2025

B.E., Electronic Information Science and Technology

GPA:3.7/4.0

➤ **Core courses:**

Intelligent Systems and Robotics / Data and Algorithm / Embodied Artificial / Probability and Stochastic Processes / Intelligent UAV Design and Implementation

➤ **Bachelor Thesis:** Robot Planning and Control based on Deep Learning (Thesis defense grade: 4.0/4.0)

University of Pennsylvania, PA, United States

09/2025-06/2027

M.E., Robotics

GPA:3.7/4.0

➤ **Core courses:**

Feedback Control / Machine Perception / Engineering Mathematics / Learning And Control For Adaptive And Reactive Robots

Publication

Jianing Qian, **Qinhe Peng**, Emmanuel Panov, Leonor Fermoselle, Dinesh Jayaraman, Bernadette Bucher, and Tarik Kelestemur. "Scaling Spatial and Temporal Context for Robotic Imitation Learning Policies With Scene Graphs", CVPR 2026.

Dongyun Zou, Zhuoyang Zhang, Junyu Chen, Wenkun He, **Qinhe Peng**, Hanrong Ye, Yao Lu, Hongxu Yin, Yu Wang, Song Han, and Han Cai. "JetViT: Efficient High-Resolution Vision Transformer with Post-Training Attention Search", CVPR Findings 2026.

Yang Yuanming, **Qinhe Peng**, Xinxing Yuan, Panyu Hou, Binxiang Qi, and Luming Duan. "A Closed-Loop Control System with Ultrafast Monitoring Based on a Photoelectric Sensor." In 2024 IEEE SENSORS, pp. 1-4. IEEE, 2024(Oral).

Selected Awards and Honors

Beijing Natural Science Foundation Funding	2025
BiRen Technology GPU Programming Competition Winner	2024
Science and Technology Innovation Excellent Scholarship, Tsinghua University	2024/2023
Student Research Training Outstanding Project Reward (top 2% of school), Tsinghua University	2022
Innovation Talent Training Program, Tsinghua University	2022

Research Experiences

Scene Graph Enabled Robot Imitation Learning

09/2025-11/2025

GRASP LAB, University of Pennsylvania / Supervised by Prof. Dinesh Jayaraman

- Enabled robots to reason over long-horizon, multi-step tasks by leveraging scene information.
- Proposed a dynamic, object-centric, scene-graph-based structured memory for imitation learning to handle partial, changing observability in large-scale environments.
- Improved policy performance of Diffusion Policy in both simulated mobile manipulation and real-world tabletop manipulation settings.
- Demonstrated stronger generalization and robustness under severe partial observability compared to baseline approaches.

Portrait Video Generation with Optical Flow and Identification

07/2025-08/2025

Peking University / Supervised by Prof. Shuai Yang

- Enabled Portrait Video Generation with Optical Flow and Identification to enhance the consistency and reasonableness of the generated video.
- Cleaned and processed over 1,000,000 videos, extracted poses and text description as generation prompts.
- Helped setting up the finetune pipeline using Megatron.

EfficientViT for Medical Images Segmentation and Efficient Video Segmentation

12/2023-Present

MIT Han Lab / Supervised by Prof. Song Han

- Proposed an efficient medical image segmentation model to improve energy efficiency while overcoming SOTA accuracy, in order to improve real-time medical image analysis in resource-constrained environments.
- Participated in MICCAI E2MIP challenge and currently adapting the model for video segmentation tasks.

- Reduced the computing cost and increased the computing speed significantly: Cutting the #MACs by **a half**, reducing #parameters to **below than 1/8** of the original, scaling up the throughput to **more than twice** its original level.
- Compared our model to formal leading model on validation dataset and zeroshot dataset. Based on the same finetune dataset, our model significantly outperformed formal leading model in every situation: on 5 different zeroshot tests our model led by **5%~29%** dices score.
- Conducted efficient versions of video and concept segmentation based on SAM2 and SAM3, ongoing.

PID High-precision Laser Signal Control System

09/2023-12/2024

Center for Quantum Information, Tsinghua University/Supervised by Prof. Luming Duan

- Applied neural network algorithms on high-precision laser control systems to stabilize frequency and amplitude.
- Simulated the effect of different PID algorithm on optical system with MATLAB.
- Carried out experiments on optical experiments, FPGA hardware and PID algorithm.
- Built up UDP communication, Wave Displayer, Frequency Analysis on PC, Arbitrary Waveform Generator on FPGA.
- Led the Beijing Natural Science Foundation Funding project. Work accepted as an oral presentation at IEEE SENSORS 2024.

Robotic Affordance Based on Diffusion

07/2023-08/2023

MMLAB, HongKong University/ Supervised by Prof. Ping Luo,

- Aimed to deal with the limitation in robotic affordance(e.g. the difficulty in complex, continuous or colaboratvie afforance analysis).
- Accomplished thorough literature review in robotic affordance, diffusion model and their related area, diving into various types and functions of robotic affordance.
- Designed independently the network, which carried out reinforcement learning in affordance with VLM, LLM, Diffusion Model and Operating Model.

Super Resolution Digital-Physical Twin

06/2023-12/2023

MIT Quantum Photonics & AI Group / Supervised by Sophia Duan

- Produced an accurate noise model of EMCCD camera by experiment, modeling and simulation.
- Conducted a set of Python code to modeling the noise of EMCCD camera.
- Accomplished the calibration and evaluation of camera distortion based on the specific EMCCD experimental image.

Robust Design Method for On-Chip Optical System

11/2022-03/2023

Tsinghua University / Supervised by research scientist Jiyuan Zheng

- Increased the robustness of the Optical System design and thus enhance chip's ability of enduring fabrication error.
- Introduced the idea of generalization in neural network into on-chip optical system design and fabrication(e.g., diffraction neural network adder chip).
- Produced an effective robust design method and its complementing code with reserved interfaces to be used in different situations.

Selected Projects

Stereo Camera Disparity Estimation & Object Detection on Gigapixel-Level Human Centric Video Dataset

- Stereo Camera 3D print, calibration in Matlab, and depth detect.
- Object detection on the large-scale multi-object dataset with YOLO, DETR, and innovative enhancement.

Fire Fighting Unmanned Drone & Auto Freight Vehicle

- UAV to automatically search and detect the fire areas and rescue targets: ROS for control, learning method for visual and detection, control theory for flight control.
- Automatic following and routing vehicle: Arduino for control, IoT for communication, SolidWorks for structure design.

Additional Information

- **Teaching Assistant:** Data and Algorithm 09/2023-07/2024, Applied Machine Learning 01/2026-Present
- Skills: Python, C/C++, Linux, Matlab, Verilog&FPGA, PyTorch, Arduino, ROS, Robot Platforms(e.g.Droid)
- Languages: English (Fluent) (GRE331+3.5, TOEFL111, Scores from first attempt)